Third-Order Polynomial Approximation of a Nonlinear Function

Abstract

A nonlinear function was approximated using a third-order polynomial regression model. Least-squares fitting was applied and performance evaluated using RMSE and R^2.

Methodology

Original function: $y = \sin(x) + 0.1x^2$ over the interval [0,10].

Fitted polynomial: $y = -0.00877x^3 + 0.26701x^2 + -0.89584x + 1.27683$

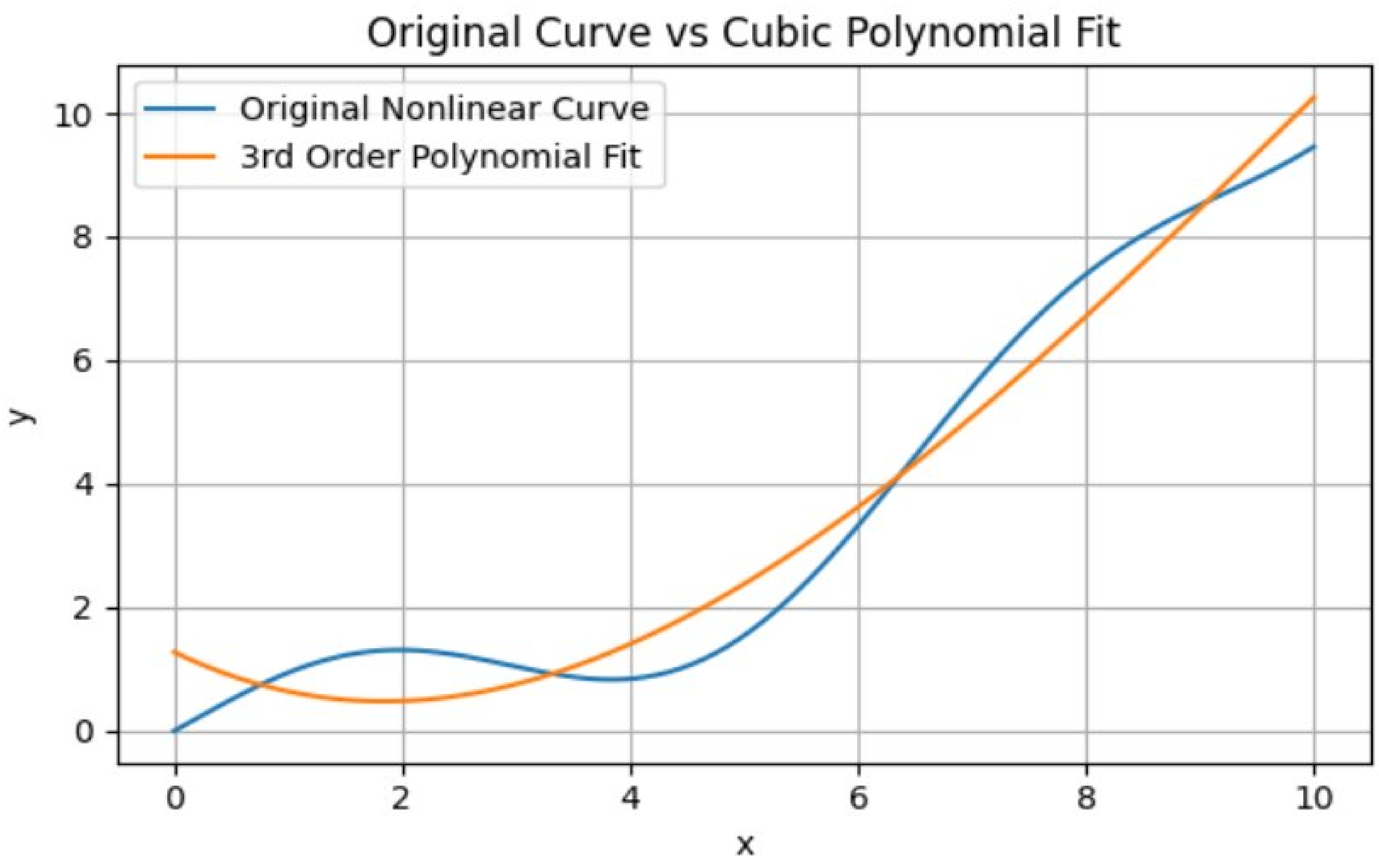

Figure 1. Original nonlinear curve versus cubic polynomial fit.

Results

RMSE = 0.5807

R^2 = 0.9650

Detailed Numerical Results

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x=0, Actual=0.000, Fit=1.277, Error=1.277
x=1, Actual=0.941, Fit=0.639, Error=0.302
x=2, Actual=1.309, Fit=0.483, Error=0.826
x=3, Actual=1.041, Fit=0.756, Error=0.285
x=4, Actual=0.843, Fit=1.405, Error=0.561
x=5, Actual=1.541, Fit=2.377, Error=0.836
x=6, Actual=3.321, Fit=3.621, Error=0.300
x=7, Actual=5.557, Fit=5.083, Error=0.474
x=8, Actual=7.389, Fit=6.711, Error=0.679
x=9, Actual=8.512, Fit=8.452, Error=0.060
x=10, Actual=9.456, Fit=10.254, Error=0.798
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Conclusion

The cubic polynomial captured the overall trend of the nonlinear function with high goodness of fit. Small deviations remained due to the oscillatory nature of the original function.

References

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2. Hastie, T., Tibshirani, R., Friedman, J. The Elements of Statistical Learning. Springer.

3. Press, W.H. et al. Numerical Recipes: The Art of Scientific Computing. Cambridge University Press.